

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
Student Name:	K Navya	Reg. No.:	192511231
Section / Batch:	CSE 2 nd Year	Date:	22/07/2026

Code of Conduct

I K Navya(192511231)_____

(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K Navya_____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25

Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	



SIMATS Online Programming Lab

DS PROGRAMMING



★ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Your Fixed Code

```

1 #include <stdio.h>
2
3 int binarySearch(int arr[], int n, int key)
4 {
5     int low = 0;
6     int high = n - 1;
7
8     while (low <= high)
9     {
10         int mid = low + (high - low) / 2;
11
12         if (arr[mid] == key)
13         {
14             return mid;
15         }
16         else if (arr[mid] < key)
17         {
18             low = mid + 1;
19         }
20         else
21         {
22             high = mid - 1;
23         }
24     }
25     return -1;
26

```

Test Input

stdin input (optional)

Output

Element found at index 5

Pending manual review



SIMATS Online Programming Lab

DS PROGRAMMING



★ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Your Fixed Code

```

1 #include <stdio.h>
2
3 #define MAX 100
4
5 int stack[MAX];
6 int top = -1;
7
8 int pop()
9 {
10     if (top == -1)
11     {
12         printf("Stack Underflow\n");
13         return -1;
14     }
15     else
16     {
17         return stack[top--];
18     }
19 }
20
21 int main()
22 {
23     // Example: Push some elements
24     stack[++top] = 10;
25     stack[++top] = 20;
26     stack[++top] = 30;
27
28     printf("Popped element: %d\n", pop());
29

```

Test Input

stdin input (optional)

Output

Popped element: 30
Popped element: 20
Popped element: 10
Stack Underflow
Popped element: -1

Pending manual review



Your online security may be at risk for scams!

McAfee can automatically identify risky texts, emails, and deepfake videos so you can stay ahead of potential fraud and misinformation.

Reactivate McAfee

 **SIMATS**
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING

KOPPALA NAVYA
192511231RB

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top;
7
8 void init()
9 {
10     top = -1; // Correct initialization
11 }
12
13 int main()
14 {
15     init();
16     printf("Top = %d\n", top);
17     return 0;
18 }
```

Submitted

Test Input

stdin input (optional)

Output

Top = -1

Pending manual review

 **McAfee**

Your online security may be at risk for scams!

McAfee can automatically identify risky texts, emails, and deepfake videos so you can stay ahead of potential fraud and misinformation.

Reactivate McAfee

 **SIMATS**
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING

KOPPALA NAVYA
192511231RB

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 char stack[100];
4 char postfix[100];
5 int top = -1, j = 0;
6
7 void push(char ch)
8 {
9     stack[++top] = ch;
10 }
11
12 char pop()
13 {
14     if (top == -1)
15         return '\0';
16     return stack[top--];
17 }
18
19 int main()
20 {
21     char ch = ' ';
22
23     /* Example stack contents: (A+B */
24     push('(');
```

Test Input

stdin input (optional)

Output

Postfix: +

 **McAfee**

Your online security may be at risk for scams!

McAfee can automatically identify risky texts, emails, and deepfake videos so you can stay ahead of potential fraud and misinformation.

Reactivate McAfee



SIMATS
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING

 KOPPALA NAVYA
192511231RB

Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 char stack[MAX];
6 int top = -1;
7
8 void push(char ch)
9 {
10     if (top < MAX - 1)
11         stack[++top] = ch;
12 }
13
14 char pop()
15 {
16     if (top == -1)
17         return '\0';
18     return stack[top--];
19 }
20
21 int checkBracket(char ch)
22 {
23     if (top == -1)
24         return 0;
25
26     if ((ch == '(' && stack[top] == '(') ||
```

Test Input

stdin input (optional)

Output

simatslab - Search

SIMATS Online Lab

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=355



SIMATS
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING

 KOPPALA NAVYA
192511231RB

5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Q10

temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 struct Node
4 {
5     int data;
6     struct Node *next;
7 };
8
9 void display(struct Node *head)
10 {
11     struct Node *temp = head;
12
13     while (temp != NULL)
14     {
15         printf("%d ", temp->data);
16         temp = temp->next;
17     }
18 }
```

Run Save Submit Fix

Test Input

stdin input (optional)

Output

simatslab - Search SIMATS Online Lab

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=355

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

KOPPALA NAVYA 192511231RB

5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack 5 marks

Q6 Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List Data Structures 5 marks

Q9 Linked List 5 marks

Q10

else:
return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int stack[MAX];
6 int top = -1;
7
8 int pop()
9 {
10     if (top == -1)
11     {
12         printf("Stack Underflow\n");
13         return -1;
14     }
15     else
16     {
17         return stack[top--];
18     }
19 }
20
21 int main()
```

Test Input

stdin input (optional)

Output

Popped: 20
Popped: 10
Stack Under
Popped: -1

Pending m

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Mark-up and share

09:22 10-08-2026

simatslab - Search SIMATS Online Lab

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=355

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

KOPPALA NAVYA 192511231RB

5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack 5 marks

Q6 Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List Data Structures 5 marks

Q9 Linked List 5 marks

Q10

current = current->next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node
5 {
6     int data;
7     struct Node *next;
8 };
9
10 struct Node* reverse(struct Node *head)
11 {
12     struct Node *prev = NULL;
13     struct Node *current = head;
14     struct Node *next = NULL;
15
16     while (current != NULL)
17     {
18         next = current->next; // Save next node
19         current->next = prev; // Reverse link
20         prev = current; // Move prev
21         current = next; // Move current
22     }
23 }
```

Test Input

stdin input (optional)

Output

Compilation
C:/CCPPComp
x86_64-w64-
mingw32/10.
-w64-mingw
C:/ccppcom

Pending m

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Mark-up and share

09:22 10-08-2026

simatslab - Search SIMATS Online Lab

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=355

SIMATS ENGINEERING **SIMATS Online Programming Lab** DS PROGRAMMING KOPPALA NAVYA 192511231RB

Stack 5 marks

Q6 Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List Data Structures 5 marks

Q9 Linked List 5 marks

Q10 Queue 5 marks

10/10 answered

```
7 struct Node *next;
8 };
9
10 struct Node *head = NULL;
11
12 void insertAtBeginning(int value)
13 {
14     struct Node *newNode;
15
16     newNode = (struct Node *)malloc(sizeof(struct Node));
17     newNode->data = value;
18
19     newNode->next = head;
20     head = newNode;
21 }
22
23 int main()
24 {
25     insertAtBeginning(10);
26     insertAtBeginning(20);
27
28     struct Node *temp = head;
29     while (temp != NULL)
30     {
31         printf("%d ", temp->data);
32         temp = temp->next;
33     }
34
35     return 0;

```

20 10

Pending manual review

09:23 10-08-2026

simatslab - Search SIMATS Online Lab

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=355

SIMATS ENGINEERING **SIMATS Online Programming Lab** DS PROGRAMMING KOPPALA NAVYA 192511231RB

Q5 Stack 5 marks

Q6 Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List Data Structures 5 marks

Q9 Linked List 5 marks

Q10 Queue 5 marks

10/10 answered

```
2
3 if front == -1:
4     print("Underflow")
5
6 else:
7     item = Q[front]
8
9     if front == rear:
10        front = -1
11        rear = -1
12    else:
13        front = front + 1
14
15    return item

```

Submitted

Output

Compilation Error:
C:\Windows\TEMP\sandbox_192511231RB_6a74230cd9dd56.00515010\main.c :1:1: warning: return type defaults to 'int' [-Wimplicit-int]

Pending manual review

© 2026 SIMATS Online Lab. All rights reserved.

09:23 10-08-2026

simatslab - Search SIMATS Online Lab

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=355

SIMATS ENGINEERING **SIMATS Online Programming Lab** DS PROGRAMMING KOPPALA NAVYA 192511231RB

Q5 Stack 5 marks ✓
Q6 Linked List 5 marks ✓
Q7 Stack 5 marks ✓
Q8 Linked List Data Structures 5 marks ✓
Q9 Linked List 5 marks ✓
Q10 Queue 5 marks ✓
10/10 answered

```
2  
3 if front == -1:  
4     print("Underflow")  
5  
6 else:  
7     item = Q[front]  
8  
9     if front == rear:  
10        front = -1  
11        rear = -1  
12    else:  
13        front = front + 1  
14  
15    return item
```

Submitted

Output

```
Compilation Error:  
C:\Windows\TEMP\sandbox_192511231  
RB_6a74230cd9dd56.00515010\main.c  
:1:1: warning: return type  
defaults to 'int' [-Wimplicit-  
int]
```

Pending manual review

© 2026 SIMATS Online Lab. All rights reserved.

simatslab - Search SIMATS Online Lab

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=356

SIMATS ENGINEERING **SIMATS Online Programming Lab** DS PROGRAMMING KOPPALA NAVYA 192511231RB

Q4 Queue 5 marks ✓
Q5 Queue 5 marks ✓
Q6 Doubly Linked List 5 marks ✓
Q7 Stack 5 marks ✓
Q8 Stack 5 marks ✓
Q9 Circular Queue 5 marks ✓
Q10 Stack 5 marks ✓
10/10 answered

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 100  
4  
5 int stack[MAX], top = -1, size;  
6  
7 void push(int item)  
8 {  
9     if(top == size - 1)  
10    {  
11        printf("Stack Overflow\n");  
12        return;  
13    }  
14    stack[++top] = item;  
15 }  
16  
17 void pop()  
18 {  
19     if(top == -1)  
20     {  
21         printf("Stack Underflow\n");  
22         return;  
23     }
```

Test Input

stdin input (optional)

Output

```
Security Error: Disallowed code  
pattern detected.
```

Pending manual review

simatslab - Search

SIMATS Online Lab

+

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=356

Chat



SIMATS
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING

KOPPALA NAVYA
192511231RB

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Q9
Circular Queue
5 marks

Q10
Stack
5 marks

}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[], int l, int r, int x) {
2     if (l > r)
3         return -1;
4
5     int mid = l + (r - l) / 2;
6
7     if (arr[mid] == x)
8         return mid;
9
10    if (arr[mid] > x)
11        return binarySearch(arr, l, mid - 1, x);
12
13    return binarySearch(arr, mid + 1, r, x);
14 }
```

Test Input

stdin input (optional)

Output

Compilation Error:
C:/CCPPComp/CCPP2/bin/./lib/gcc/
x86_64-w64-
mingw32/10.3.0/./../x86_64-
-w64-mingw32/bin/ld.exe:
C:/ccppcomp/ccpp2/bin/./lib/gcc/
x86_64-w64-mingw32/bin/ld.exe: cannot find -lmingw32

Pending manual review

Submitted

Windows taskbar with search bar, task view, and various application icons. System tray shows language (ENG IN), network, and time (09:31 10-08-2026).

simatslab - Search

SIMATS Online Lab

+

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=356

Chat



SIMATS
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING

KOPPALA NAVYA
192511231RB

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Q9
Circular Queue
5 marks

Q10
Stack
5 marks

}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void insertFront(int data) {
2     if (front == 0)
3         front = MAX - 1;
4     else
5         front = front - 1;
6
7     deque[front] = data;
8 }
```

Test Input

stdin input (optional)

Output

Compilation Error:
C:\Windows\TEMP\sandbox_192511231-
RB_6a76ee39bee675.37745397\main.c
: In function 'insertFront':
C:\Windows\TEMP\sandbox_192511231-
RB_6a76ee39bee675.37745397\main.c

Pending manual review

Submitted

Windows taskbar with search bar, task view, and various application icons. System tray shows language (ENG IN), network, and time (09:31 10-08-2026).

simatslab - Search SIMATS Online Lab

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=356

SIMATS ENGINEERING **SIMATS Online Programming Lab** DS PROGRAMMING KOPPALA NAVYA 192511231RB

Binary Search 5 marks

Q4 Queue 5 marks

Q5 Queue 5 marks

Q6 Doubly Linked List 5 marks

Q7 Stack 5 marks

Q8 Stack 5 marks

Q9 Circular Queue 5 marks

Q10 Stack 5 marks

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void enqueue(int data) {
2     if (rear >= MAX)
3         return;
4
5     queue[rear++] = data;
6 }
```

Test Input

stdin input (optional)

Output

Pending manual review

Submitted

Windows taskbar: Search, File Explorer, Edge, VS Code, etc. 09:31 10-08-2026

simatslab - Search SIMATS Online Lab

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=356

SIMATS ENGINEERING **SIMATS Online Programming Lab** DS PROGRAMMING KOPPALA NAVYA 192511231RB

Q5 Queue 5 marks

Q6 Doubly Linked List 5 marks

Q7 Stack 5 marks

Q8 Stack 5 marks

Q9 Circular Queue 5 marks

Q10 Stack 5 marks

10/10 answered

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *prev;
7     struct Node *next;
8 };
9
10 void insertEnd(struct Node **head, int data) {
11     struct Node *newNode = malloc(sizeof(struct Node));
12
13     newNode->data = data;
14     newNode->next = NULL;
15     newNode->prev = NULL;
16
17     if (*head == NULL) {
18         *head = newNode;
19         return;
20     }
21
22     struct Node *temp = *head;
23
24     while (temp->next != NULL)
25         temp = temp->next;
26
27     temp->next = newNode;
28     newNode->prev = temp;
29 }
```

Test Input

stdin input (optional)

Output

Pending manual review

Windows taskbar: Search, File Explorer, Edge, VS Code, etc. 09:31 10-08-2026

simatslab - Search SIMATS Online Lab

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=356

SIMATS ENGINEERING **SIMATS Online Programming Lab** DS PROGRAMMING KOPPALA NAVYA 192511231RB

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 Identify the issue in the below pseudocode. POP():
2
3 if top == 0 -> Underflow
```

Test Input

stdin input (optional)

Output

Pending manual review

Submitted

Q3 Binary Search 5 marks ✓
Q4 Queue 5 marks ✓
Q5 Queue 5 marks ✓
Q6 Doubly Linked List 5 marks ✓
Q7 Stack 5 marks ✓
Q8 Stack 5 marks ✓
Q9 Circular Queue 5 marks ✓
Q10

09:32 10-08-2026

simatslab - Search SIMATS Online Lab

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=356

SIMATS ENGINEERING **SIMATS Online Programming Lab** DS PROGRAMMING KOPPALA NAVYA 192511231RB

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 Point out the error.
2
3 if top == MAX
4
5 overflow
```

Test Input

stdin input (optional)

Output

Pending manual review

Submitted

Q2 Circular Queue 5 marks ✓
Q3 Binary Search 5 marks ✓
Q4 Queue 5 marks ✓
Q5 Queue 5 marks ✓
Q6 Doubly Linked List 5 marks ✓
Q7 Stack 5 marks ✓
Q8 Stack 5 marks ✓
Q9 Circular Queue 5 marks ✓

09:32 10-08-2026

simatslab - Search

SIMATS Online Lab

+

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=356

Chat



SIMATS
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING



KOPPALA NAVYA
192511231RB

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Q9
Circular Queue
5 marks

Q10
Stack
5 marks

✱ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if ((rear + 1) % MAX == front) {
2     printf("Queue Full");
3     return;
4 }
5
6 rear = (rear + 1) % MAX;
7 queue[rear] = vehicle;
```

Submitted

Test Input

stdin input (optional)

Output

Pending manual review

Search

ENG IN

09:32 10-08-2026

simatslab - Search

SIMATS Online Lab

+

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=356

Chat



SIMATS
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING



KOPPALA NAVYA
192511231RB

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Q9
Circular Queue
5 marks

Q10
Stack
5 marks

10/10 answered

```
6 int top = -1;
7
8 // Push an element into the stack
9 void push(int data)
10 {
11     if (top == MAX - 1)
12     {
13         printf("\nStack Overflow!");
14         return;
15     }
16     top++;
17     stack[top] = data;
18     printf("\n%d pushed into the stack.", data);
19 }
20
21 // Pop an element from the stack
22 void pop()
23 {
24     if (top == -1)
25     {
26         printf("\nStack Underflow!");
27         return;
28     }
29     printf("\n%d popped from the stack.", stack[top]);
30     top--;
31 }
```

Output

Pending manual review

Search

ENG IN

09:32 10-08-2026

